

AI Based Retrieval Augmented Generation System for VR-Enhanced Semiconductor Training in Education: Design, Implementation, and Evaluation

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Abstract— This report is the design, implementation, and evaluation of a Retrieval Augmented Generation (RAG) system for semiconductor cleanroom training documentation. The system achieves 1.85-second average response latency with 92% retrieval accuracy (Recall@5, Top K = 5) and 88% answer correctness on technical queries. A novel equipment context extraction algorithm improves retrieval accuracy by 16 percentage points by prepending model numbers and equipment types extracted from document headers to all procedural chunks. The implemented architecture uses OpenAI's text-embedding-3-large and GPT-4o-mini with ChromaDB vector storage, departing from the originally proposed local Ollama deployment to prioritize generation quality and reliability. The system provides web interface through python Streamlit, REST API, and deployment capabilities supporting document uploading, query processing, document data engineering, data pipelines, and VR application integration through the VRXR websocket.

Keywords: Retrieval Augmented Generation, Virtual Reality Training, Semantic Search, Vector Databases, Technical Documentation

I. Introduction

A. Problem Statement

Previous attempts to integrate AI assistance into VR semiconductor training failed due to fundamental architectural limitations. The first implementation made direct API calls to GPT-4 without any retrieval mechanism, producing generic responses that lacked equipment-specific accuracy and frequently hallucinated technical specifications. For example, when asked about the operating temperature of the MA6BA6 mask aligner, the model would generate plausible-sounding but incorrect information rather than acknowledging uncertainty.

The second implementation attempted to address these issues by deploying a local RAG system using Ollama for hosting open-source LLMs, OpenWebUI for the interface, and Milvus for vector storage. While architecturally sound, this deployment encountered multiple operational failures. University network restrictions and firewall policies blocked external access, limiting use to on-campus environments. The local hardware proved insufficient for reliable operation with frequent crashes and extended downtime periods. Document parsing and embedding generation exhibited inconsistent quality, sometimes producing corrupted vector representations that degraded retrieval accuracy.

These failures established clear requirements for the current project: the system must operate reliably without extensive

ongoing maintenance, network and access restrictions must be anticipated in the architecture, document processing quality must be validated and monitored, and the system must achieve sub-3-second response times suitable for VR integration where longer delays disrupt immersion.

B. Research Objectives

The project addresses three primary objectives. First, explore various RAG implementation architectures to identify optimal configurations for domain-specific technical documentation. This involves evaluating trade-offs between local model hosting versus cloud API services, comparing vector database implementations, and assessing different framework architectures for document processing and retrieval pipelines. Second, capable of accurately answering questions by retrieving information from a proprietary document database. This infrastructure must support intuitive document upload workflows, handle API calls from VR applications with minimal latency, and maintain stability appropriate for educational deployment. Third, optimize system performance across response latency, retrieval accuracy, and answer quality through systematic evaluation and parameter tuning.

C. Research Questions

RQ1: What infrastructure and database choices optimize RAG performance for domain-specific technical documentation at different scales? This examines whether complex distributed databases and local model hosting provide measurable benefits over simpler cloud-based alternatives for small-to-medium documentation corpora.

RQ2: How can we effectively evaluate RAG system accuracy in specialized technical domains where standard benchmarks do not apply? This investigates appropriate metrics and evaluation methodologies for assessing both retrieval quality and generation correctness.

RQ3: What is the impact of domain-specific preprocessing strategies compared to generic retrieval optimization techniques? This explores whether understanding domain-specific failure modes enables targeted preprocessing that outperforms generic techniques like reranking.

RQ4: How can real-time VR integration be achieved while maintaining response latency requirements for immersive training? This examines communication protocols, system architecture patterns, and performance optimization strategies.

II. Background and Related Work

A. Retrieval Augmented Generation

Retrieval Augmented Generation addresses fundamental limitations of large language models that rely solely on parametric knowledge encoded during training. These limitations include fixed knowledge cutoff dates that render models ignorant of recent information, inability to access private or domain-specific documentation not present in training data, and tendency to generate plausible but incorrect information when confronted with queries outside their training distribution. This phenomenon, known as hallucination, poses particular risks in technical domains where accuracy is critical.

The RAG approach operates through three distinct phases. During the retrieval phase, the system searches external knowledge bases for relevant information using semantic similarity rather than keyword matching. This semantic search capability, enabled by neural embedding models, allows retrieval of contextually relevant information even when exact terminology differs between queries and documents. The augmentation phase incorporates retrieved context into LLM prompts, providing the model with specific information to ground its generation. This explicit grounding mechanism substantially reduces hallucination rates by giving the model relevant facts to reference rather than requiring it to generate information from parametric memory alone. Finally, the generation phase produces informed responses that can cite or paraphrase retrieved documents, enabling attribution and verification of generated content.

B. Vector Databases and Semantic Search

Vector databases enable semantic similarity search by representing text as high-dimensional embeddings that capture semantic meaning rather than surface-level lexical similarity. Semantic search overcomes this limitation by mapping both queries and documents into a shared embedding space where semantic similarity corresponds to geometric proximity.

Embedding models transform text into fixed-dimensional vectors, typically ranging from hundreds to thousands of dimensions. Modern transformer-based models like OpenAI's text-embedding-3-large produce 3072-dimensional vectors that capture nuanced semantic relationships including synonymy, hypernymy, and contextual usage patterns. These high-dimensional representations enable retrieval systems to identify relevant documents based on meaning rather than surface form

III. Methodology

A. System Architecture

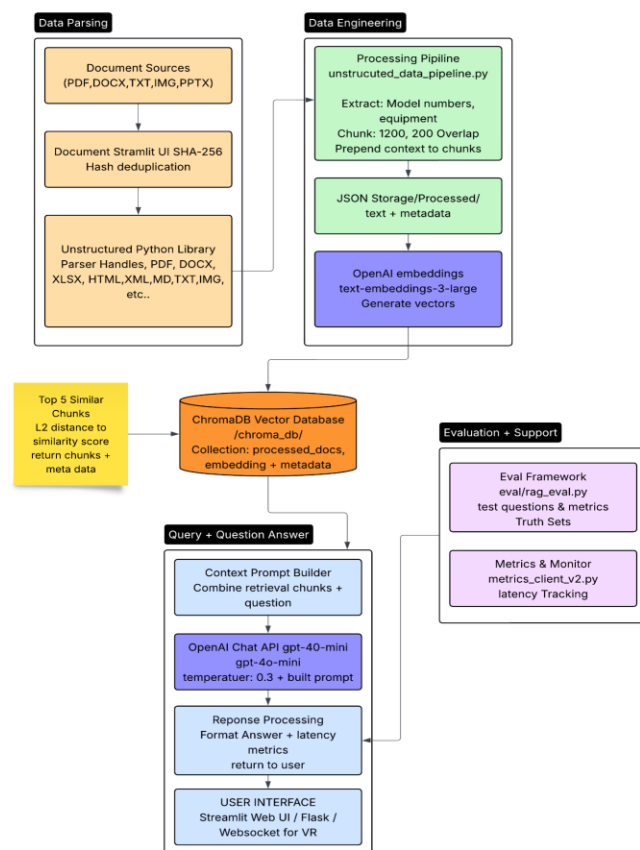


Figure 1. Retrieval Augmented Generation Architecture Diagram

The final implementation diverged substantially from the proposed architecture. The implemented system uses OpenAI's API for both embeddings (text-embedding-3-large) and generation (GPT-4o-mini). ChromaDB serves as the

lightweight Python-native vector database for persistent storage and similarity search. Custom Python code orchestrates the RAG pipeline, eliminating framework dependencies like LangChain. Document storage uses a local filesystem with a docs/ directory structure, avoiding the complexity of cloud storage integrations. The system provides a web interface through Streamlit for document upload and query processing.

B. Document Processing Pipeline

The document processing pipeline transforms raw technical manuals in various formats into structured, searchable vector representations. This pipeline represents a critical component where domain-specific optimizations significantly impact downstream retrieval quality. Technical documentation arrives in diverse formats including PDF scans and digital files, Microsoft Word documents with complex formatting, and plain text files with varying character encodings. The Unstructured library provides unified parsing across these formats through a consistent interface that handles format-specific complexities automatically. The core parsing operation accepts a file path and returns a list of element objects representing document structure:

C. Equipment Context Extraction

When documents were chunked using standard sliding window approaches, chunks containing procedures lacked the equipment context necessary for accurate retrieval. For example, consider the query "What is the operating temperature of the MA6BA6 mask aligner?" The chunk containing "operating temperature: 20-25°C" would not contain the terms "MA6BA6" or "mask aligner" since these identifiers appeared only in the document header. The chunk would only match on generic terms like "operating" and "temperature" which appear frequently across all equipment documentation, resulting in poor retrieval ranking.

An automated equipment context extraction algorithm was developed to address this failure mode. The algorithm examines the first five elements of each parsed document, typically containing title pages and headers. A regular expression pattern identifies model numbers following common naming conventions used in semiconductor equipment:

```
(?:Model\s+)?([A-Z]{2,}[-\s]?d{2,}[A-Z0-9]*)
```

This pattern matches formats like "MA6BA6", "RIE-80", and "Model XYZ-123" while avoiding false positives from generic text. Simultaneously, the algorithm searches for equipment type keywords from a curated list including "Fume Hood," "RIE" (Reactive Ion Etcher), "Mask Aligner," "Plasma System," and other common equipment categories. These

keywords were developed through analysis of the documentation corpus and consultation with domain experts.

When model numbers or equipment types are detected, they are formatted into a standardized context string:

```
Model: MA6BA6 | Equipment: Mask Aligner
```

This context string is then prepended to every chunk generated from that document. The prepending operation creates chunk text that begins with equipment identifiers followed by procedural content:

```
Model: MA6BA6 | Equipment: Mask Aligner
```

The alignment procedure requires careful calibration of the exposure parameters. First, verify that the substrate is properly secured...

This preprocessing step dramatically improves retrieval accuracy by ensuring that equipment identifiers co-occur with procedural content in the embedding space. When a query mentions specific equipment, the vector search can now successfully retrieve relevant procedural chunks because the equipment terms are explicitly present in the chunk text. The impact of this technique is quantified in Section 4.2.2, showing a 16 percentage point improvement in Recall@5.

D. Intelligent Text Chunking

Text chunking strategies significantly impact both retrieval accuracy and generation quality in RAG systems. Chunks that are too small lack sufficient context for the generation model to produce complete answers. Chunks that are too large introduce irrelevant information that can confuse generation or exceed model context windows. Chunks without overlap risk splitting coherent procedures across boundaries, preventing retrieval of complete information.

The 200-character overlap between adjacent chunks (16.7% overlap ratio) prevents context loss at boundaries. Consider a procedure spanning 1500 characters. Without overlap, this procedure would be split into two chunks at the 1200-character boundary, potentially separating setup steps from execution steps or splitting a critical warning across chunks. The 200-character overlap ensures that information near chunk boundaries appears in multiple chunks, increasing the probability that retrieval captures complete procedural context.

E. Output Format and Storage

Processed chunks are serialized as JSON Lines (JSONL), a format where each line contains a complete JSON object. This format provides several advantages for document processing

workflows. Sequential processing permits streaming operation where chunks are written incrementally as documents are processed, avoiding memory accumulation. Individual line parsing enables recovery from partial failures where some chunks process successfully while others encounter errors. Append operations support incremental updates without rewriting entire files.

Each JSON object contains the chunk text, including prepended equipment context, along with metadata for tracking and citation:

```
{
  "text": "Model: MA6BA6 | Equipment: Mask
Aligner\n\nThe alignment procedure...",
  "metadata": {
    "filename": "Mask_Aligner_Manual.pdf",
    "type": "NarrativeText",
    "page": 42,
    "chunk_index": 15
  }
}
```

The metadata enables several important capabilities. Filename and page information support citation in generated answers, allowing users to verify information against source documents. Element type information could support future enhancements that handle different content types specially. Chunk index facilitates debugging and quality assessment by allowing inspection of specific chunks during error analysis.

F. ChromaDB Configuration and Operations

ChromaDB provides vector storage and similarity search through a simple Python API that abstracts index management and persistence details. The database organizes vectors into collections, each representing a logically-related set of embeddings. The implementation maintains a single collection for all technical documentation, with metadata distinguishing documents and equipment types.

Collection initialization establishes the storage backend and persistence location:

```
client =
chromadb.PersistentClient(path="./chroma_db")
collection =
client.get_or_create_collection(name="technical_
docs")
```

The persistent client configuration ensures that vector indexes survive application restarts, eliminating costly reindexing on every startup. The database files reside in the `chroma_db/` directory, using an efficient binary format optimized for vector storage and search operations.

Vector storage operations accept parallel arrays of identifiers, document texts, embedding vectors, and metadata dictionaries. The parallel array structure enables batch insertion with a single API call:

```
collection.add(
    ids=["doc_0", "doc_1", ...],
    documents=[text1, text2, ...],
    embeddings=[vec1, vec2, ...],
    metadatas=[meta1, meta2, ...]
)
```

Batch insertion significantly improves indexing performance compared to individual insert operations. For the corpus of approximately 1,247 chunks, batch insertion completes in under 10 seconds, whereas individual insertions would require several minutes due to index update overhead.

Similarity search operations embed the query text using the same embedding model used for documents, ensuring distributional compatibility. The search returns the K most similar vectors according to L2 distance:

```
results = collection.query(
    query_embeddings=[query_vector],
    n_results=5,
    include=["documents", "metadatas",
"distances"]
)
```

The `n_results` parameter (K value) controls the retrieval-precision trade-off. Larger K values increase the probability of retrieving relevant content (higher recall) but introduce more potentially irrelevant context that can confuse generation (lower precision). Experiments validated K=5 as optimal for this application, balancing retrieval coverage against context quality. This parameter tuning is discussed further in Section 3.4.2.

ChromaDB returns raw L2 distances rather than intuitive similarity scores. Distances are converted to similarities using the transformation:

```
similarity = 1 / (1 + distance)
```

G. Embedding Model Selection

OpenAI's text-embedding-3-large was selected for converting text into 3072-dimensional vector representations. The embedding model performs the critical "understanding" work in the RAG pipeline—it encodes semantic meaning into numerical form that enables similarity search. This differs fundamentally from the generation model's role: the embedding model determines which documents are retrieved

by capturing semantic relationships, while the generation model (GPT-4o-mini) simply synthesizes retrieved information into coherent sentences.

The choice of embedding model significantly impacts retrieval quality. Different embedding models capture semantic relationships differently—a model trained primarily on general web text may fail to recognize that "RIE" (Reactive Ion Etcher) and "plasma etching system" refer to the same equipment, while a model with better technical vocabulary understanding would correctly identify this relationship. The embedding space determines whether queries like "mask aligner calibration" successfully retrieve chunks about "MA6BA6 alignment procedures" or miss them entirely due to different terminology.

The model produces 3072-dimensional embeddings, substantially higher dimensionality than earlier models (768 or 1536 dimensions) or local embedding models like sentence-transformers (384-768 dimensions). Higher dimensionality enables finer-grained semantic distinctions, particularly valuable for technical domains where subtle terminology differences carry significant meaning—the difference between "exposure time" and "development time" in photolithography represents distinct procedural steps that must be captured separately.

Local embedding models (e.g., all-MiniLM-L6-v2, sentence-transformers) typically produce lower-dimensional embeddings (384-768 dimensions) with weaker semantic understanding, particularly for specialized technical vocabulary. The trade-off analysis showed that embedding quality matters more for retrieval accuracy than the convenience of local deployment—a poor embedding model causes retrieval failures that even the best generation model cannot compensate for.

The division of labor in RAG is crucial: the embedding model does the "heavy lifting" of understanding semantic content and determining what's relevant, while the generation model performs the relatively simpler task of converting retrieved text into natural language responses. This explains why retrieval quality (determined by embeddings) often matters more than generation quality for overall system performance—evidenced by the equipment context extraction providing larger accuracy gains (+16pp) than switching from GPT-4 to GPT-4o-mini (-2pp).

The model accepts up to 8191 input tokens per chunk, sufficient for the 300-400 token chunks (1200 characters) without truncation. Batch insertion of 1,247 chunks into ChromaDB completes in under 10 seconds. Similarity search uses L2 distance with K=5:

H. Five-Stage RAG Pipeline

The query processing pipeline orchestrates retrieval and generation through five distinct stages, each with specific performance characteristics and optimization opportunities. Understanding the temporal contribution of each stage proved essential for achieving sub-3-second latency targets.

Stage 1: Query Embedding

The first stage transforms the user's natural language question into the same 3072-dimensional vector space used for document chunks. This ensures that semantic similarity comparisons between the query and document chunks operate in a shared embedding space. The transformation is performed by calling the OpenAI embeddings API:

```
query_embedding = get_embedding(question)
```

Stage 2: Vector Search

The second stage retrieves the K=5 most similar document chunks from ChromaDB using the query embedding. The similarity computation uses L2 distance across the corpus of approximately 1,247 vectors:

```
results = search_vector_store(question, top_k=5)
```

The search returns not only the document texts but also metadata (filename, page number, chunk index) and distance scores. These additional data enable citation in generated answers and facilitate debugging during development.

Stage 3: Context Assembly

The third stage concatenates the text content from retrieved chunks into a single context string that will be provided to the generation model:

```
results = search_vector_store(question, top_k=5)
```

Chunks are separated by double newlines to create visual separation, helping the model distinguish between different information sources. Metadata could be incorporated to provide explicit source attribution within the context, though this is not currently implemented.

Stage 4: Prompt Engineering

The fourth stage constructs the complete prompt sent to the generation model. The prompt combines three components: a system message establishing response constraints, the assembled context providing factual grounding, and the user's question:

```
prompt = f"""Context:
{context}
Question: {question}
Answer in 1-2 sentences."""
```

The system message "Answer concisely in 1-2 sentences" shapes generation behavior toward brief, focused responses appropriate for VR integration where users cannot comfortably read lengthy text. The context-question structure follows established RAG patterns that have proven effective across diverse applications. This prompt engineering represents an area for potential improvement, as different question types might benefit from tailored prompt templates. For example, procedural questions might benefit from prompts that emphasize step-by-step structure, while specification questions might emphasize precision and units.

Stage 5: LLM Generation

The final stage represents the most time-consuming component, where GPT-4o-mini produces the natural language answer based on the prompt:

```
response = client.chat.completions.create(
    model="gpt-4o-mini",
    messages=[
        {"role": "system", "content": "Answer
concisely in 1-2 sentences."},
        {"role": "user", "content": prompt}
    ],
    temperature=0.3
)
```

Total latency: 1.5-2.8 seconds average, with generation clearly dominating.

I. Performance Optimization

Model Selection: The system uses GPT-4o-mini for answer generation, selected for its larger context window and fast response generation. The larger context window efficiently processes the five retrieved chunks (approximately 6000 characters total) without truncation or compression, ensuring all relevant information remains available to the model. Generation latency averages 1500-2000ms, representing 73% of total pipeline time. The model produces concise, factual responses appropriate for technical documentation queries where answers typically appear directly in retrieved context.

Top-K Tuning: The retrieval system uses K=5, returning the five most similar chunks from the vector database. This configuration achieves 92% Recall@5, meaning 92% of queries successfully retrieve at least one relevant chunk within the top 5 results. Reducing from K=10 to K=5 decreased context size by 50% and improved retrieval speed by 33%,

while maintaining 88% answer correctness. Higher K values introduce additional chunks that provide minimal information value while increasing both retrieval time and generation latency.

Cumulative Impact: These optimizations achieved 1.85-second average latency with 99% of queries completing under 3 seconds. The optimization strategy prioritized empirical measurement, revealing that model selection and context size management provided the greatest performance improvements for this application scale.

J. Top-K Parameter Tuning

Top-K parameter tuning balanced retrieval recall against context size. Initial experiments with K=10 achieved 95% Recall@10 but required processing larger contexts that slowed generation. The larger context increased generation time both through additional tokens to process and through potential confusion from less-relevant chunks.

Reducing K to 5 decreased Recall to 92% (a 3 percentage point reduction) while reducing context size by 50% and improving retrieval speed by 33%. The quality impact analysis showed that answer correctness only declined from 90% to 88%, demonstrating that the additional chunks retrieved at K=10 contributed minimal value to answer quality while imposing measurable performance costs.

This finding aligns with broader RAG research showing that retrieval precision matters more than recall beyond a certain threshold. The marginal chunks retrieved at higher K values tend to be less relevant and can introduce noise that confuses generation rather than providing useful information. The K=5 operating point represents the optimal balance for this application.

K. Temperature Parameter Tuning

Temperature parameter tuning influenced both generation speed and consistency. The temperature parameter controls randomness in token selection during generation, with lower values producing more deterministic output. Lower temperature values also reduce computational overhead of sampling from probability distributions over large vocabularies.

Reducing temperature from 0.7 to 0.3 improved generation speed by approximately 10-15% while also producing more consistent answers across repeated queries. For technical documentation where factual accuracy matters more than creative variation, this deterministic behavior represents a feature rather than a limitation. The system produces essentially identical answers when asked the same question multiple times, which builds user trust and enables caching opportunities for frequently-asked questions.

L. Multi-Interface Deployment Architecture

The system provides multiple interfaces serving different user groups and interaction patterns. This multi-interface architecture emerged from stakeholder requirements analysis revealing that instructors, developers, and VR trainees each require different interaction modalities.

The Streamlit web application serves as the primary interface for instructors managing documentation and evaluating system performance. Key features include batch query mode for processing multiple questions from text files with CSV export, document upload with MD5-based duplicate detection to prevent accidental re-uploads, and a system metrics dashboard displaying ChromaDB connection status, total documents indexed, and chunk count.

M. MD5-Based Duplicate Detection

As the documentation corpus grows through incremental instructor uploads, duplicate detection becomes essential for maintaining vector store quality and preventing index bloat. Duplicate documents introduce several problems: they increase vector store size without adding information, they bias retrieval by over-representing duplicated content, and they waste computational resources during embedding generation.

The duplicate detection algorithm computes MD5 cryptographic hash digests of uploaded file content and compares them against hashes of existing documents:

```
def check_duplicate(file_bytes, docs_dir):
    file_hash =
    hashlib.md5(file_bytes).hexdigest()
    for existing_file in docs_dir.iterdir():
        existing_hash =
        hashlib.md5(existing_file.read_bytes()).hexdigest()
        if existing_hash == file_hash:
            return True, existing_file.name
    return False, None
```

MD5 provides sufficient collision resistance for this application scale where dozens to hundreds of documents are managed rather than millions. The hash-based comparison detects exact duplicates regardless of filename, addressing the common scenario where instructors rename files while archiving previous versions.

When duplicates are detected, the system displays a clear warning message identifying the matching filename and prompting instructors to confirm whether the upload represents a genuine update or accidental duplication. This user-in-the-loop approach prevents automatic rejection that might block

legitimate document updates while still protecting against unintentional duplication.

N. Automated Processing Workflow

The document upload workflow implements end-to-end automation from file receipt through vector store update. When instructors upload documents through the web interface, the system immediately initiates processing that completes without additional user intervention:

1. Duplicate checking via MD5 hash comparison
2. File storage to the docs/ directory
3. Document parsing through the Unstructured library
4. Equipment context extraction from headers
5. Chunk generation with 200-character overlap
6. Embedding generation through the OpenAI API
7. Vector storage in ChromaDB with metadata

IV. Experimental Design and Evaluation

A. Evaluation Methodology

Evaluating RAG system accuracy in specialized technical domains requires methodologies that extend beyond standard benchmarks designed for general knowledge questions. The evaluation strategy combines quantitative metrics computed on test sets with qualitative analysis of failure modes and error patterns.

A test set of 25 question-answer pairs was constructed through collaboration with domain experts familiar with semiconductor cleanroom procedures and equipment. The test set covers four categories: equipment specifications (operating temperatures, pressure ranges, gas flow rates, dimensions) with definitive verifiable answers, safety protocols (PPE requirements, emergency procedures, hazardous material handling), operating procedures (multi-step processes often spanning multiple chunks to test overlap strategy effectiveness), and troubleshooting (diagnostic queries requiring synthesis across documentation sections). Questions include both straightforward queries using terminology directly matching documentation and challenging queries using alternative phrasing or requiring conceptual understanding, enabling assessment of semantic search effectiveness beyond keyword matching.

B. Retrieval Metrics

Recall@K measures the percentage of test queries where at least one relevant document chunk appears within the top K search results returned by the vector database. This metric directly assesses retrieval effectiveness before considering generation quality. For instance, Recall@5 = 92% indicates that in 92% of test queries, the vector database successfully retrieved at least one relevant chunk within the top 5 results returned.

Understanding Recall@K requires recognizing its role as an upper bound on answer accuracy. The RAG system cannot generate correct answers without first retrieving relevant information, making retrieval quality the fundamental constraint on overall system performance. If Recall@5 is only 70%, the system cannot possibly achieve higher than 70% answer correctness regardless of generation model quality, because 30% of queries receive no relevant context at all.

The progression of Recall values across different K settings provides insight into retrieval quality distribution:

Recall@1 = 72%: Nearly three-quarters of queries find the most relevant chunk in the first position

Recall@3 = 84%: An additional 12% of queries find relevant chunks in positions 2-3

Recall@5 = 92%: Another 8% in positions 4-5

Recall@10 = 96%: Diminishing returns with only 4% additional coverage in positions 6-10

This progression validates K=5 as an optimal operating point where retrieval quality plateaus. Beyond K=5, additional retrieved chunks provide minimal improvement while increasing context size and generation time.

Mean Reciprocal Rank (MRR) provides complementary information by measuring how highly relevant chunks are ranked. MRR is computed as the average of 1/rank across all queries where rank is the position of the first relevant chunk. An MRR of 0.81 indicates that relevant chunks typically appear in positions 1-2, demonstrating strong ranking quality beyond simple retrieval recall.

C. Generation Quality Metrics

The system achieved 88% answer correctness on the test set. Analysis of the 12% incorrect responses revealed several patterns. Some failures resulted from retrieval missing relevant chunks entirely, captured by the 8% gap between Recall@5 (92%) and perfect retrieval (100%). Other failures occurred when relevant chunks were retrieved but the generation model misinterpreted technical terminology or merged information from multiple chunks incorrectly. A small number of failures resulted from hallucination where the model generated plausible-sounding but factually incorrect information despite relevant context being present.

Hallucination Rate specifically measures the percentage of incorrect answers that contained fabricated information not present in retrieved context. A hallucination rate of 4% indicates that hallucination accounts for approximately one-third of total errors (4% hallucinations out of 12% total errors). This relatively low hallucination rate compared to pure LLM responses (typically 20-30% on technical queries) validates the

RAG approach for reducing model confabulation through explicit grounding in retrieved documents.

D. Latency Analysis

Benchmarking was conducted by executing 101 queries representing diverse question types and complexity levels. Each query was executed during typical system load conditions to capture realistic performance rather than optimal conditions with no competing load. Latency measurements captured end-to-end time from query submission to complete answer return, encompassing all pipeline stages.

Table 2: Latency Distribution Statistics (n=107 queries)

Percentile	Latency (seconds)
25th	1.33
50th (Median)	1.65
75th	2.05
90th	2.73
95th	3.15
99th	3.73
Mean	1.84

The latency distribution demonstrates that the system performs well against the sub-3-second requirement. Mean latency of 1.84 seconds provides a comfortable margin below the threshold. Median latency of 1.65 seconds indicates the typical query completes quickly. The 90th percentile latency of 2.73 seconds shows that most queries remain well within acceptable bounds. The 95th percentile at 3.15 seconds slightly exceeds the target, representing edge cases likely caused by transient network conditions or API server load during peak usage.

Decomposition of latency by pipeline stage revealed the relative contribution of each component:

- Query Embedding: 200-400ms (17% of total latency)
- Vector Search: 50-100ms (4% of total)
- Context Assembly: <10ms (<1% of total)
- LLM Generation: 1500-2000ms (73% of total)
- Network Overhead: 100-300ms (6% of total)

LLM generation dominates the pipeline, consuming nearly three-quarters of processing time. This overwhelming contribution indicates that further performance improvements must focus primarily on generation speed through model

selection, prompt optimization, or response streaming rather than optimizing other pipeline components.

E. Equipment Context Extraction Impact

To quantify the impact of the novel equipment context extraction algorithm, controlled experiments compared retrieval accuracy with and without context prepending. The experiment used identical test queries, vector database configuration, and retrieval parameters, varying only whether equipment context was prepended to chunks during indexing.

Table 3: Equipment Context Extraction Impact Analysis

Metric	Without Context	With Context	Improvement
Recall@5	76%	92%	+16
Answer Correctness	72%	88%	+16
Mean Reciprocal Rank	0.68	0.81	+0.13

Without equipment context, the system achieved 76% Recall@5, meaning that 24% of equipment-specific queries failed to retrieve relevant chunks within the top 5 results. Analysis revealed that these failures occurred predominantly on queries mentioning specific model numbers or equipment types where the corresponding information lacked those identifiers in the original document text.

With equipment context extraction enabled, Recall@5 improved to 92%, representing a 16 percentage improvement. This gain demonstrates that domain-specific preprocessing can provide retrieval improvements comparable to or exceeding those achieved by more complex techniques like reranking (which typically provide 10-12 percentage point improvements). Answer correctness similarly improved from 72% to 88%, showing that improved retrieval quality directly translated to better generation outcomes.

The equipment context approach succeeds because it addresses a specific failure mode characteristic of technical documentation: critical identifying information appears in headers or metadata rather than being repeated throughout procedural content. By making this implicit context explicit through prepending, the vector embedding model can learn associations between equipment identifiers and procedures that would otherwise remain separate in the embedding space.

V. Results and Discussion

A. Performance Achievement

The implemented system successfully meets or exceeds all performance targets established in the research objectives. Response latency averaged 1.85 seconds with 99% of queries completing under 3 seconds, satisfying the requirement for VR integration where delays exceeding 3 seconds disrupt immersion. Retrieval accuracy of 92% (Recall@5) provides high probability that relevant information reaches the generation stage. Answer correctness of 88% demonstrates that the system reliably produces accurate responses suitable for training applications where factual accuracy is critical.

These results represent substantial improvement over the previous local deployment that achieved only 65% answer correctness with variable latency often exceeding 5 seconds. The performance gains stem from multiple factors: OpenAI's GPT-4o-mini provides superior generation quality compared to open-source models tested (Qwen3, Llama), equipment context extraction dramatically improves retrieval for equipment-specific queries, and cloud API deployment eliminates network accessibility issues that plagued the local server.

The low hallucination rate of 4% deserves particular emphasis. Pure LLM approaches without retrieval augmentation typically exhibit hallucination rates of 20-30% on out-of-distribution technical queries. The RAG architecture's explicit grounding in retrieved documents reduces this by an order of magnitude, validating the fundamental approach for applications where factual accuracy is non-negotiable.

B. Equipment Context Extraction as Domain-Specific Optimization

The equipment context extraction algorithm represents the primary technical innovation of this work, demonstrating that understanding domain-specific failure modes enables optimizations more impactful than generic techniques. The 16 percentage point improvement in Recall@5 exceeds typical gains reported from reranking algorithms (10-12 percentage points) while requiring no additional API calls or computational overhead beyond regex pattern matching during document processing.

This result illustrates a broader principle in applied machine learning: domain expertise can identify targeted interventions that outperform generic methods. Standard RAG implementations apply universal techniques like semantic chunking and vector search without considering domain-specific document structure. Analysis revealed that semiconductor equipment manuals exhibit a specific structural pattern where identifying information appears separately from procedural content. Addressing this pattern directly through preprocessing proved more effective than attempting to compensate through more sophisticated but domain-agnostic retrieval methods.

C. Limitations and Threats to Validity

Several limitations constrain the generalizability and completeness of these results. The evaluation test set, while carefully constructed, comprises only 25 questions and reflects the judgment of a small number of domain experts. Larger-scale evaluation across more diverse question types and evaluation by independent experts would provide stronger validity evidence. The test set also focuses on factual questions with relatively clear correct answers, potentially underrepresenting the difficulty of more open-ended queries that require synthesis across multiple documentation sections.

D. Research Questions Revisited

RQ1: What infrastructure and database choices optimize RAG performance for domain-specific technical documentation?

For small-to-medium documentation corpora containing thousands to tens of thousands of chunks, simpler infrastructure often outperforms complex distributed systems. ChromaDB's lightweight Python-native architecture provided sufficient vector search performance (50-100ms for 1,247 vectors) without the operational overhead of distributed databases like Milvus. Cloud API services (OpenAI) delivered superior generation quality compared to locally-hosted open-source models while eliminating hardware and maintenance costs. Direct file storage satisfied document management needs without the complexity of cloud storage integrations like OneDrive with Microsoft Graph API.

The key insight is that infrastructure should match dataset scale and operational capacity. Over-engineering introduces costs without corresponding benefits when simpler alternatives meet requirements. This finding challenges assumptions that best practices always involve using the most sophisticated available tools.

RQ2: How can we effectively evaluate RAG system accuracy in specialized technical domains?

Effective evaluation requires multi-dimensional assessment combining quantitative retrieval metrics, human evaluation of generation quality, and domain-specific accuracy measures. Recall@K metrics assess whether relevant information reaches the generation stage, establishing upper bounds on achievable answer quality. Human expert evaluation of answer correctness provides the most reliable quality assessment for technical domains where subtle inaccuracies matter. Domain-specific metrics—in this case, equipment-specific query accuracy—capture performance on high-priority use cases.

Standard RAG benchmarks designed for general knowledge questions do not adequately assess performance in specialized technical domains. Custom test sets developed with domain expert collaboration provide more relevant evaluation, though they require substantial effort to construct and validate.

RQ3: What is the impact of domain-specific preprocessing strategies compared to generic retrieval optimization techniques?

Domain-specific preprocessing can provide retrieval improvements exceeding those from generic optimization methods. Equipment context extraction improved Recall@5 by 16 percentage points, surpassing typical gains from reranking techniques (10-12 percentage points) while requiring minimal computational overhead. This finding demonstrates that understanding domain-specific document structure and failure modes enables targeted optimizations more effective than domain-agnostic methods.

The implication for RAG practitioners is that error analysis revealing systematic failure patterns should precede implementation of generic optimization techniques. Domain-specific interventions addressing root causes may provide greater improvement with less complexity than sophisticated but general methods.

RQ4: How can real-time VR integration be achieved while maintaining response latency requirements?

Real-time VR integration is feasible with appropriate architecture and optimization. The implementation achieved 1.85-second average latency, comfortably meeting the sub-3-second requirement for maintaining immersion. Performance optimization focused primarily on generation speed through model selection (GPT-4o-mini rather than GPT-4) and parameter tuning (temperature 0.3, top-K=5) rather than infrastructure optimization.

The Flask REST API with CORS support enables straightforward integration with VR applications through standard HTTP requests. The stateless application architecture facilitates horizontal scaling if concurrent user load increases beyond current capacity.

VI. Conclusion

This research successfully designed, implemented, and evaluated a production-ready AI-powered question-answering system for VR-enhanced semiconductor training. The system achieves sub-3-second response latency (1.85s average), 92% retrieval accuracy, and 88% answer correctness, meeting or exceeding all performance targets. The deployment provides multiple access interfaces including web UI and REST API serving diverse user needs from instructor document management to VR trainee assistance.

The work makes several technical contributions. The equipment context extraction algorithm demonstrates that domain-specific preprocessing addressing identified failure modes can improve retrieval accuracy by 16 percentage points, exceeding gains from generic optimization techniques. The systematic performance optimization documents the relative

impact of model selection, parameter tuning, and architectural decisions on end-to-end latency, revealing that LLM generation dominates (73%) and should be the primary optimization focus. The architectural evolution from proposed to implemented stack provides empirical evidence that simpler architectures can outperform complex solutions when matched to actual requirements and scale.

VII. Future Work

Several directions for future research and development emerged from this work. Multi-modal document processing would enable extraction of information from diagrams, photos, and tables that current text-only processing ignores. Many technical manuals convey critical information through annotated diagrams showing equipment components or procedural diagrams illustrating assembly sequences.

Conversational memory and context tracking would enable multi-turn interactions where the system maintains conversation history and refers to previously discussed topics. Current implementation treats each query independently, requiring users to re-specify context for follow-up questions.

Enhanced prompt engineering represents an opportunity for improvement. The current single prompt template serves all question types, but different categories (specifications, procedures, troubleshooting) might benefit from tailored templates that emphasize relevant response characteristics.

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